Individual Research Gap Analysis

1. SWE-InfraBench

Research Gaps

The primary research gap identified from SWE-InfraBench is the absence of an integrated validation framework that supports multiple IaC formats beyond AWS CDK, combines static analysis with live deployment validation, enforces comprehensive security compliance, monitors operational behavior post-deployment, and scales to enterprise-level smart manufacturing cloud environments. There is a clear need for a framework that can evaluate LLM-generated IaC not just on syntactic correctness and unit test passing, but on actual deployability, security posture, and operational reliability. Such a framework would need to incorporate deployment engines, security scanners, and monitoring tools into a unified pipeline that provides actionable feedback to LLM agents for iterative refinement, thereby bridging the gap between benchmark performance and production-ready infrastructure code.

2. Multi-IaC-Bench

Research Gaps

The overarching research gap identified from Multi-IaC-Bench is the need for a more realistic evaluation framework that combines static analysis with deployment-time validation, security compliance checking, and operational monitoring in actual cloud environments, rather than relying on synthetic data and static assessment. There is a clear requirement for a framework that can ingest real-world user requests, generate and mutate IaC across multiple formats, validate deployability through live provisioning, enforce security policies, and provide iterative feedback for refinement. Such a framework would need to address the limitations of LLM judge bias by incorporating human validation or more objective deployment-based metrics, and it would need to scale to the complexity and diversity of enterprise manufacturing cloud infrastructures where multi-cloud strategies are increasingly common.

3. IaCGen (Deployability-Centric Infrastructure-as-Code Generation: Fail, Learn, Refine, and Succeed through LLM-Empowered DevOps Simulation)

Research Gaps

The research gaps identified from IaCGen center on the need for a fully autonomous, cost-effective, and security-aware refinement framework that minimizes human intervention while maximizing deployment success, security compliance, and user-intent alignment.

There is a clear requirement for intelligent simulation or predictive techniques that can reduce the number of costly deployment attempts by identifying potential failures before actual provisioning. Security policies need to be integrated from the first iteration rather than applied as an afterthought, ensuring that generated IaC is secure by design. Intent modeling and interactive clarification mechanisms are needed to improve alignment between generated code and user requirements. The framework must also be extended to support multi-cloud and hybrid environments, and post-deployment capabilities such as drift detection and automated remediation must be incorporated to ensure ongoing infrastructure reliability. Ultimately, the goal is to create a system that can autonomously generate, validate, refine, deploy, and monitor IaC with minimal human involvement, which is particularly valuable for smart manufacturing where speed, reliability, and security are equally critical.

4. RIVA (Reliable Configuration Drift Detection using LLM Agents)

Research Gaps

The primary research gap identified from RIVA is the need for an end-to-end framework that integrates drift detection as one component of a unified infrastructure validation and remediation pipeline, connecting generation, deployment validation, security checking, drift detection, and automated repair into a cohesive system. Drift detection must be validated in real cloud environments with production-like workloads, and it must be correlated with security and compliance policies to provide actionable, prioritized alerts. Most importantly, detected drift should automatically trigger corrective IaC generation and deployment, closing the loop and enabling continuous compliance without human intervention. Such a framework would be particularly valuable for smart manufacturing environments where infrastructure must remain consistent and secure across distributed, multi-cloud deployments, and where any deviation from the desired state can lead to production outages or security breaches.

5. TerraProbe (Layered-Oracle Framework for Detecting Deceptive Fixes in LLM-Assisted Terraform Security Repair)

Research Gaps

The overarching research gap identified from TerraProbe is the need for a generalized framework capable of automatically generating, validating, and securely repairing Infrastructure-as-Code across multiple IaC technologies and cloud platforms, moving beyond evaluation to autonomous remediation. Such a framework would need to support multiple IaC formats, integrate with deployment engines to verify that repaired code actually provisions functional infrastructure, and incorporate comprehensive security policies to ensure that repairs are genuinely secure. The framework should also include mechanisms for intent preservation, ensuring that security repairs do not inadvertently change the intended infrastructure behavior. Furthermore, the approach must scale to enterprise environments where thousands of infrastructure modules require continuous security validation and repair. For smart manufacturing, where security vulnerabilities in infrastructure can have physical safety implications, the ability to automatically and reliably repair insecure IaC is particularly critical, and a unified framework that combines evaluation, generation, and deployment validation would represent a substantial advancement over current fragmented approaches.

Combined Research Gap Analysis

The five selected studies collectively demonstrate significant advances in LLM-assisted Infrastructure-as-Code generation, editing, validation, security repair, and configuration drift detection, yet they address these critical capabilities independently rather than as an integrated solution. SWE-InfraBench focuses exclusively on AWS CDK editing without deployment validation or security assessment, while Multi-IaC-Bench covers multiple formats but relies on synthetic data and static evaluation without live deployment testing. IaCGen emphasizes deployability through iterative refinement but suffers from high operational costs, requires human intervention, and achieves only 8.4 percent security compliance and 25.2 percent user-intent alignment. RIVA introduces innovative multi-agent drift detection but concentrates solely on drift without integration with generation, validation, or automated remediation pipelines. TerraProbe provides rigorous security repair evaluation but is limited to Terraform and emphasizes detection over automatic secure repair generation. Common limitations across all papers include restricted support for specific IaC technologies, reliance on synthetic or benchmark datasets, limited deployment-time validation in live cloud environments, insufficient security assurance and compliance enforcement, continued dependence on human intervention, and the absence of post-deployment operational monitoring and drift remediation. These observations collectively reveal the critical need for an intelligent, scalable, and unified Infrastructure-as-Code validation framework that supports multiple IaC formats, performs real-world deployment validation, enforces comprehensive security compliance, detects and remediates configuration drift, and provides automated LLM-driven recommendations with minimal human involvement. Such a framework would be particularly transformative for complex smart manufacturing cloud environments where reliability, security, automation, and operational continuity are equally paramount, bridging the gap between fragmented academic benchmarks and the holistic, production-ready infrastructure management required by modern enterprises.